

Changyi Li

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Education

Shanghai Innovation Institute (SII) & Fudan University

Ph.D. in Artificial Intelligence

Sep 2026 - Jun 2029 (Expected)

Shanghai, China

– Advised by Asst. Prof. [Xudong Pan](#) and Prof. [Min Yang](#); co-advised by Prof.

[Xipeng Qiu](#).

Fudan University, System Software and Security Lab

M.S. in Computer Science; GPA: 3.96/4.00, Rank: 1/379

Sep 2024 - Jun 2026

Shanghai, China

– Advisors: Asst. Prof. [Xudong Pan](#) and Prof. [Min Yang](#).

Qilu University of Technology (Shandong Academy of Sciences)

B.S. in Artificial Intelligence; GPA: 4.50/5.00, Rank: 1/102

Sep 2020 - Jun 2024

Shandong, China

Research Interests

My research focuses on **safety**, **alignment**, and **capability development** for **frontier AI systems**. I work on scalable oversight, risk evaluation, and recursive self-improvement in agentic systems, while also exploring embodied intelligence.

Selected Honors and Awards

First-Class Scholarship, Fudan University

2026

Freshman Scholarship, Fudan University

2025

Principal Scholarship (1/38,000+)

2024

Provincial Outstanding Graduate (Top 0.1%)

2024

National Scholarship (Top 0.1%)

2023

MCM/ICM Meritorious Winner, Team Leader

2023

First-Class Scholarship & Merit Student Pacesetter

2022, 2023

Publications

[1] **AutoControl Arena: Synthesizing Executable Test Environments for Frontier AI Risk Evaluation.**

Changyi Li, Pengfei Lu, Xudong Pan, Fazl Barez, Min Yang

ICML, 2026. [Page] [Paper] [Code] [Machine Heart] [PaperWeekly]

[2] **ReasoningShield: Safety Detection over Reasoning Traces of Large Reasoning Models.**

Changyi Li, Jiayi Wang, Xudong Pan, Geng Hong, Min Yang

Under review at NeurIPS, 2026. Reviews: 5/5/4 (all positive). [Page] [Paper] [Code] [Models] [Dataset]

[3] **CyberEvolver: Structured Self-Evolution for Cybersecurity Agents On the Fly.**

Yihe Fan, **Changyi Li**, Lichen Xu, Xudong Pan, Jiarun Dai, Hong Geng, Min Yang

Under review at NeurIPS, 2026. [Page] [Paper] [Code]

[4] **Large Language Model-Powered AI Systems Achieve Self-Replication with No Human Intervention.**

Xudong Pan, Jiarun Dai, Yihe Fan, Minyuan Luo, **Changyi Li**, Min Yang

arXiv, 2025. [Page] [Paper] [LiveScience] [Forbes] [The Independent]

Research Experience

AutoControl Arena: Automated Frontier AI Risk Evaluation Platform

First Author | Jul 2025 – Mar 2026

- **Motivation:** Autonomous agents greatly expand the risk surface of AI systems, creating long-tail frontier risks; manual benchmarks are costly and hard to scale, while pure LLM simulators lack causal realism.
- **Method:** Built an Architect–Coder–Monitor framework: Architect structures natural-language risks, Coder synthesizes executable sandboxes via logic–narrative decoupling, and Monitor audits trajectory-level risks from behavior and reasoning traces.
- **Contribution:** Created AutoControl Arena for end-to-end automated risk evaluation; released X-Bench with 70 scenarios across 15 domains and 7 risk categories; introduced Stress–Temptation testing for long-tail frontier risks.
- **Results:** Achieved **98%** end-to-end construction success; reproduced OpenAI/Anthropic high-risk behaviors and identified alignment illusion, bidirectional safety scaling, and stress–temptation coupling effects.
- **Impact:** Received **100+ GitHub stars**, **600+ clones**, and media coverage from *Machine Heart* and *PaperWeekly*; delivered an invited talk at the **Seoul Alignment Workshop**.

ReasoningShield: CoT Content Safety Auditing for Reasoning Models

First Author | Feb 2025 – Jun 2025

- **Motivation:** CoT reasoning introduces hidden safety risks: harmful content can appear in long intermediate reasoning even when the final answer looks harmless or the model later refuses.
- **Method:** Built a 3-level, 10-category CoT risk taxonomy and trained **ReasoningShield-1B/3B** with SFT and DPO for harmfulness detection and step-level risk localization.
- **Contribution:** Introduced the **first CoT safety-auditing framework**, including a moderation benchmark with **9.2K** annotated query–CoT examples and a specialized moderation model for reasoning models.
- **Results:** **ReasoningShield-3B** reached SOTA on CoT safety benchmarks with **91.8%** F1, improving over strong baseline LlamaGuard-4-12B by **35.6%**; on mainstream QA safety benchmarks, it reached **85.0%** F1 (**+15.7%**).
- **Impact:** Received **5/5/4** reviews (all positive) at NeurIPS 2026; released models and data with **3K+ Hugging Face downloads**; cited by teams including **IEEE Fellow**, **IAPR Fellow**, and **National Research Council Canada**.

CyberEvolver: Self-Evolving Cybersecurity Agents

Second Author | Feb 2026 – Jun 2026

- **Motivation:** Existing cyber agents rely on fixed human-designed workflows that struggle across targets and failure modes, while generic self-evolution faces unstructured mutations, sparse and noisy execution feedback, and limited update diversity.
- **Method:** Decomposed the cybersecurity-agent Harness into four structured evolvable layers; introduced trajectory diagnosis to turn noisy execution logs into actionable mutation signals; used population-based beam search to preserve diverse agent variants.
- **Contribution:** Proposed an on-policy Harness self-evolution framework for cybersecurity tasks; on CTF, vulnerability-exploitation, and penetration-testing tasks, improved seed-agent pass@16 by **13.6%** on average and outperformed the strongest human-designed baseline by **14.0%**.

AI System Self-Replication Risk Study

Core Experimental Contributor | Nov 2024 – Feb 2025

- **Motivation:** Model self-replication is widely viewed as an AI safety red line, yet frontier AI labs at the time (e.g., OpenAI and Google) had underestimated the associated risk.
- **Method:** Designed a Perceive–Reflect–Discover–Plan harness to elicit longer-horizon self-replication beyond standard ReAct agents; evaluated 32 open-source models under a reproducible milestone-based protocol with explicit success criteria.
- **Contribution:** Provided the **first systematic empirical evidence** of end-to-end self-replication in open-source AI systems: **11/32 systems** succeeded, including a locally runnable **14B** model; further examined spontaneous replication, self-exfiltration, and shutdown evasion.
- **Impact:** Received **millions of views and reposts** on X, YouTube, and TikTok, coverage from *LiveScience*, *Forbes*,

The Guardian, and *The Independent*, plus an invited presentation at a France AI Action Summit side event.

Text-to-Image Systems Red Teaming (Alibaba Tianchi Competition)

Technical Lead | Sep 2024 – Nov 2024

- **Motivation:** Commercial text-to-image systems face full-chain content-safety and misuse risks, but many attacks target only a single model rather than the combined text-filter, generator, and image-filter stack.
- **Method:** Designed cognition-deception and semantic-association attacks with scenario rewriting, adversarial affixes, caption injection, prompt fuzzing, and VLM pre-screening.
- **Contribution:** Led full-chain gray-box and black-box vulnerability discovery across **30+** commercial systems; achieved a **90%+** text-filter bypass rate and **80%+** image attack success rate; ranked **26/1574** teams (**top 1.7%**); escalated severe vulnerabilities in **10+** mainstream commercial models to a **national security briefing**, which received central-level attention.

Industry Projects

Full-Stack Multimodal Guardrail System

Project Lead | Alibaba Cloud / CAS | Sep 2025 – Mar 2026

- **Background:** Enterprise multimodal review must cover fine-grained unsafe imagery, implicit semantic risks, and evolving compliance requirements, while real risk samples are scarce, human labels are costly, and existing pipelines are unreliable and hard to trace.
- **Method:** Built an automated data synthesis and labeling loop: constructed a risk lexicon and risk-category/concept space, expanded abstract risk terms into visual scene seeds and full image prompts, and used structured VLM review, multi-model voting, consistency scoring, and confidence calibration to produce traceable high-confidence pseudo-labels with hard-example feedback and incremental training.
- **Outcomes:** Achieved **95%+ evaluation accuracy** and **90%** efficiency improvement; supported multimodal content-safety evaluation and guardrail work for **Alibaba Cloud**, **CAS**, **China Electronics**, and **E Fund**; resulted in **one invention patent**.

AIGC Text Quality and Compliance Detection System

Project Lead | China Electronics | Jun 2025 – Sep 2025

- **Background:** China Electronics (Fortune 500) needed production-grade quality acceptance and compliance screening for AIGC artistic-text outputs, including text omissions, malformed glyphs, semantic mismatch, and sensitive content; general VLMs reached only about 60% accuracy on fine-grained defects.
- **Method:** Designed a five-stage lightweight detection pipeline: OCR and rule checks for text omissions, VLM-based image-text consistency judging, multimodal fine-tuning, ViT defect classification, and model ensembling for joint quality and compliance decisions.
- **Contribution:** Designed the defect taxonomy, constructed negative samples, trained models, and deployed the service, building a production detector across **8 problem types** and **10+ visual styles**, including typos, malformed glyphs, semantic mismatch, risky text, deformation, and aesthetic defects.
- **Outcomes:** Exceeded **95% negative-sample detection**, met a **500 ms** latency requirement, handled **500K+/day** peak volume, and delivered **10M+** high-quality data assets.

Academic Service

Invited Talk	Seoul Alignment Workshop, FAR.AI	2026
Invited Participant	Safety & Alignment closed-door mixer (ICML), OpenAI	2026
Organizer	AI Deception Workshop, Fudan & SAIF & Concordia AI [Link]	2025
Teaching Assistant	Foundation of Deep Learning, Fudan University	2025, 2026